pyAmpliCol at one loop

Benchmarks of the recursive one-loop engine against MadLoop, Recola and OpenLoops

report 2026-11-08 · newest engine measured `e852f448` · finite part, binary64, one pinned thread, $B = 128$
Table 1: engine `ca602436`, G1 2026-11-06, AMD EPYC 9754, shared host · Table 2: engine `e852f448`

×1/16 faster — ×1/4 — ×1 — ×4 — ×16 slower
run time ÷ fastest provider

Gate pass ×0.25 · fail ×1.4 · open ×1.0 · <5 ms: not gated ×1.4 — time ÷ limit
Stages inputs · numerator · scatter · NinjaRS · masters · glue
missing (why) · no colour: generation, LC · † within estimate · ¶ reference's own (DD) · ‡ user decision
ML MadLoop · Rc Recola · OL OpenLoops

Table 1: **Born-interfered one-loop processes** (the one-loop amplitude interfered with the Born). How to read it: after Table 3.

	Process · validation	Generation	Full colour · per point, batched	Leading colour (LC)
	pyAmpliCol: the most recent measured configuration — `loop` · $2 \to 2$ to $2 \to 5$: engine `ca602436`, native `loop-native-v2-slice-ca602436` (`4cbb549f`…), G1 2026-11-06 · AMD EPYC 9754, shared host			
	legs · corrections · loop families; largest deviation of the finite part per reference	generate + plan export, peak memory, plan size; provider generation on this host	median per point at $B = 128$, the gate batch; provider times; gate G1 (time ÷ limit); stage shares	batched, same configuration; LC/FC; no provider ratio
$2 \to 2$	$gg \to gg$ 4 legs · QCD · 24 families (LC 12) Rc·OL·ML ≤ 1.3e−12 16 pts · median 1.1e−14 · †	**5.70 s** ×40 0.57 GiB peak · plan 0.0240 MB **Rc 142 ms** · ML 9.70 s	**1.29 ms** ×0.64 $B = 128$ **ML 2.02 ms** · OL 3.20 ms · Rc 10.0 ms no gate (2→2) numerator 51 %	**0.596 ms** LC/FC 0.46 $B = 128$ · FC 1.29 ms
$2 \to 3$	$u\bar{u} \to d\bar{d}g$ 5 legs · QCD · 42 families (LC 27) ML·OL ≤ 7.9e−13 · Rc 3.3e−10 16 pts · median 4.8e−14 · ¶	**6.73 s** ×17 0.58 GiB peak · plan 0.0300 MB **Rc 406 ms** · ML 9.87 s	**1.72 ms** ×1.1 $B = 128$ **OL 1.52 ms** · ML 3.17 ms · Rc 3.18 ms <5 ms: not gated ×0.87 numerator 38 %	**1.12 ms** LC/FC 0.65 $B = 128$ · FC 1.72 ms
	$u\bar{u} \to u\bar{u}g$ 5 legs · QCD · 68 families (LC 42) OL·ML ≤ 1.2e−11 · Rc 7.2e−11 16 pts · median 1.4e−13 · ¶	**7.07 s** ×15 0.58 GiB peak · plan 0.0360 MB **Rc 462 ms** · ML 10.9 s	**2.64 ms** ×0.90 $B = 128$ **OL 2.92 ms** · Rc 5.55 ms · ML 8.59 ms <5 ms: not gated ×0.69 numerator 42 %	**1.59 ms** LC/FC 0.60 $B = 128$ · FC 2.64 ms
	$u\bar{u} \to e^-e^+g$ 5 legs · QCD · 11 families (LC 7) ML·Rc ≤ 7.5e−14 16 pts · median 6.8e−15	**10.1 s** ×14 0.61 GiB peak · plan 0.0530 MB **Rc 701 ms** · ML 14.0 s	**0.788 ms** ×0.83 $B = 128$ **ML 0.948 ms** · Rc 1.08 ms · OL no library <5 ms: not gated ×0.83 numerator 36 %	**0.446 ms** LC/FC 0.57 $B = 128$ · FC 0.788 ms
	$u\bar{u} \to ggg$ 5 legs · QCD · 90 families (LC 62) ML·Rc ≤ 1.5e−12 · OL 8.4e−11 16 pts · median 4.9e−14 · ¶	**6.62 s** ×10 0.58 GiB peak · plan 0.0490 MB **Rc 634 ms** · ML 17.6 s	**4.34 ms** ×0.35 $B = 128$ **OL 12.4 ms** · Rc 25.3 ms · ML 28.2 ms <5 ms: not gated ×0.27 numerator 53 %	**2.38 ms** LC/FC 0.55 $B = 128$ · FC 4.34 ms
	$gg \to ggg$ 5 legs · QCD · 154 families (LC 77) OL·Rc·ML ≤ 3.7e−7 16 pts · median 5.8e−13 · †	**8.10 s** ×1.2 0.59 GiB peak · plan 0.112 MB **Rc 6.63 s** · ML 47.0 s	**20.1 ms** ×0.21 $B = 128$ **OL 97.0 ms** · ML 247 ms · Rc 351 ms pass ×0.16 numerator 50 %	**7.59 ms** LC/FC 0.38 $B = 128$ · FC 20.1 ms
$2 \to 4$	$u\bar{u} \to e^-e^+gg$ 6 legs · QCD · 69 families (LC 33) Rc·ML ≤ 2.2e−10 16 pts · median 6.0e−14 · †	**9.53 s** ×11 0.62 GiB peak · plan 0.0680 MB **Rc 833 ms** · ML 24.8 s	**4.21 ms** ×0.41 $B = 128$ **Rc 10.2 ms** · ML 27.3 ms · OL no library <5 ms: not gated ×0.32 numerator 54 %	**1.71 ms** LC/FC 0.41 $B = 128$ · FC 4.21 ms
	$gg \to t\bar{t}gg$ 6 legs · QCD · 1 086 families (LC 375) OL·Rc·ML ≤ 8.8e−9 16 pts · median 1.2e−11 · †	**9.98 s** ×0.16 0.64 GiB peak · plan 0.320 MB **Rc 61.4 s** · ML 218 s	**79.8 ms** ×0.17 $B = 128$ **OL 472 ms** · Rc 1.97 s · ML 5.45 s pass ×0.13 numerator 44 %	**28.2 ms** LC/FC 0.35 $B = 128$ · FC 79.8 ms
	$u\bar{u} \to gggg$ 6 legs · QCD · 628 families (LC 375) Rc 8.0e−12 · OL 2.3e−10 16 pts · median 1.7e−12 · †	**9.86 s** ×0.30 0.65 GiB peak · plan 0.382 MB **Rc 32.8 s**	**65.3 ms** ×0.17 $B = 128$ **OL 374 ms** · Rc 908 ms · ML not exported pass ×0.13 numerator 57 %	**25.3 ms** LC/FC 0.39 $B = 128$ · FC 65.3 ms
	$gg \to gggg$ 6 legs · QCD · 1 100 families (LC 550) ML·Rc·OL ≤ 1.5e−7 8 pts · median 2.5e−11 · †	**14.0 s** ×0.0066 0.89 GiB peak · plan 1.23 MB **Rc 35.4 min** · ML 144.6 min	**307 ms** ×0.11 $B = 128$ **OL 2.84 s** · Rc 17.6 s · ML 38.7 s pass ×0.083 numerator 51 %	**126 ms** LC/FC 0.41 $B = 128$ · FC 307 ms

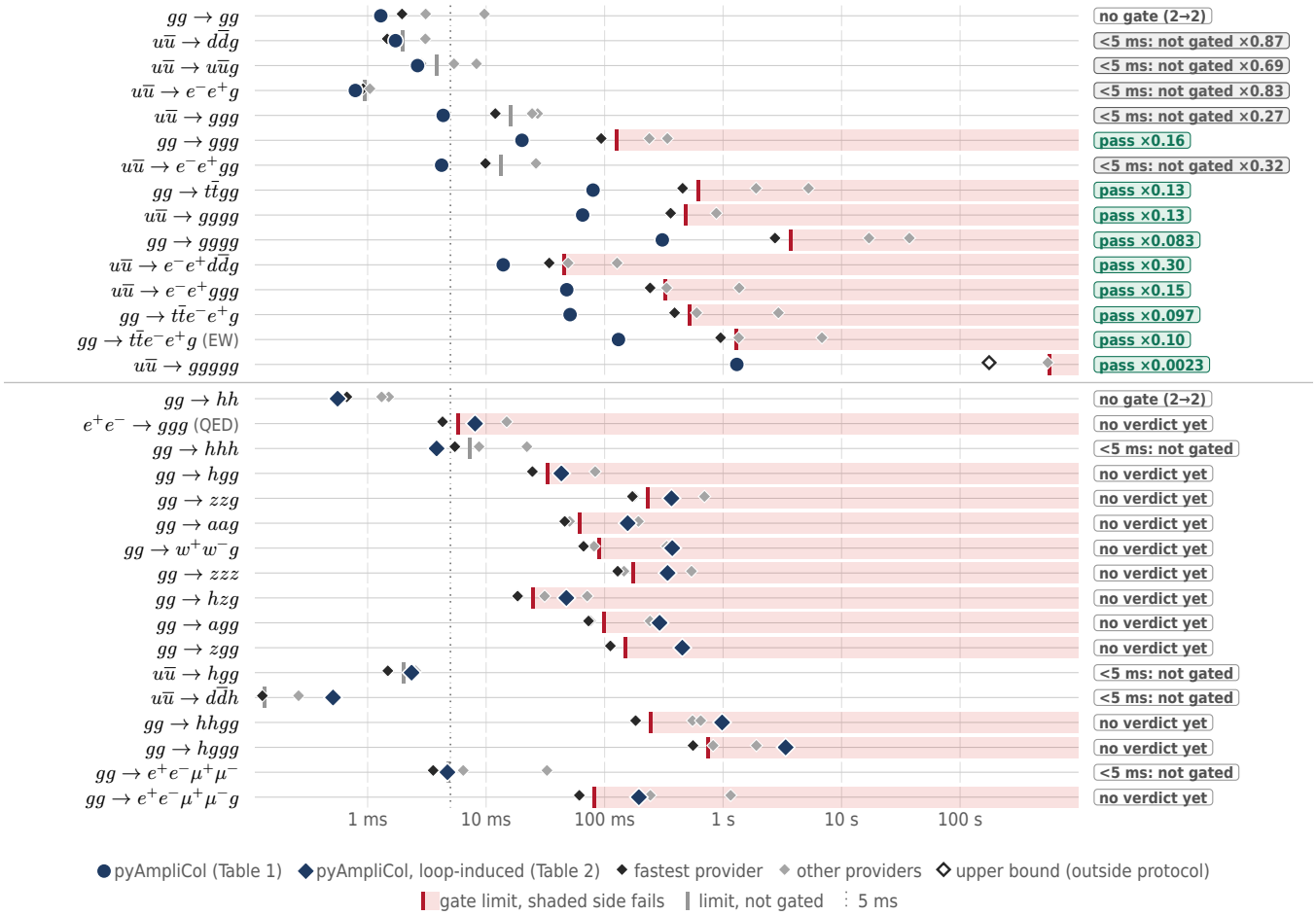
	Process · validation	Generation	Full colour · per point, batched	Leading colour (LC)
pyAmpliCol: the most recent measured configuration — loop · $2 \to 2$ to $2 \to 5$: engine ca602436, native loop-native-v2-slice-ca602436 (4cbb549f…), G1 2026-11-06 · AMD EPYC 9754, shared host				
	legs · corrections · loop families; largest deviation of the finite part per reference	generate + plan export, peak memory, plan size; provider generation on this host	median per point at $B = 128$, the gate batch; provider times; gate G1 (time ÷ limit); stage shares	batched, same configuration; LC/FC; no provider ratio
$2 \to 5$	$u\bar{u} \to e^-e^+d\bar{d}g$ 7 legs · QCD · 242 families (LC 114) OL·ML·Rc ≤ 4.5e−11 16 pts · median 5.3e−13	**12.1 s** ×1.8 0.63 GiB peak · plan 0.0960 MB **Rc 6.73 s** · ML 58.6 s	**13.9 ms** ×0.39 **Rc 35.3 ms** · OL 51.0 ms · ML 132 ms $B = 128$ pass ×0.30 numerator 48 %	**5.75 ms** LC/FC 0.41 $B = 128$ · FC 13.9 ms
	$u\bar{u} \to e^-e^+ggg$ 7 legs · QCD · 463 families (LC 164) ML·Rc·OL ≤ 4.3e−11 16 pts · median 1.4e−12 · †	**11.1 s** ×0.83 0.64 GiB peak · plan 0.179 MB **Rc 13.4 s** · ML 146 s	**47.9 ms** ×0.19 **Rc 251 ms** · OL 346 ms · ML 1.42 s $B = 128$ pass ×0.15 numerator 63 %	**9.42 ms** LC/FC 0.20 $B = 128$ · FC 47.9 ms
	$gg \to t\bar{t}e^-e^+g$ 7 legs · QCD · 579 families (LC 164) OL·Rc·ML ≤ 1.1e−8 16 pts · median 2.7e−11 · †	**11.8 s** ×0.43 0.64 GiB peak · plan 0.176 MB **Rc 27.7 s** · ML 165 s	**51.1 ms** ×0.13 **OL 404 ms** · Rc 620 ms · ML 3.04 s $B = 128$ pass ×0.097 numerator 57 %	**10.1 ms** LC/FC 0.20 $B = 128$ · FC 51.1 ms
	$gg \to t\bar{t}e^-e^+g$ 7 legs · EW · 1 599 families OL 2.7e−10 · ML 2.8e−9 16 pts · median 4.5e−10	**276 s** ×12 9.24 GiB peak · plan 0.445 MB **Rc 23.5 s** · ML 11.9 min	**131 ms** ×0.13 **Rc 987 ms** · OL 1.40 s · ML 7.07 s $B = 128$ pass ×0.10 numerator 46 % model: the workspace's py3 G_mu copy	***not supported*** refused by design: EW LC/NLC is never supported (user 2026-10-30, Q4; typed colour-declaration-coverage)
	$u\bar{u} \to ggggg$ 7 legs · QCD · 5 258 families (LC 2 712) ML 2.1e−12 · Rc 2.9e−10 3 pts · median 7.0e−11 · ¶	**88.2 s** ×0.0086 2.76 GiB peak · plan 14.7 MB **Rc 170.3 min** · ML 12.2 h	**1.31 s** ≥×0.0072 **Rc ≤182 s** · ML 569 s · OL no library $B = 128$ pass ×0.0023 numerator 62 %	**438 ms** LC/FC 0.34 $B = 128$ · FC 1.31 s
	$gg \to ggggg$ 7 legs · QCD · 4 707 families (LC 4 707) NJet DD 1.5e−7 10 pts · median 3.4e−9 · † · top-free (particle_veto = [t]), Nf = 5 ‡ pts 4, 8 above 2.0e−8: accepted (user 2026-10-25)	**133 s** 4.39 GiB peak · plan 20.2 MB ML diagrams 17.4 h, stopped in export · Rc unfinished	**4.99 s** $B = 128$ no gate (no matched provider) numerator 52 %	**2.68 s** LC/FC 0.54 $B = 128$ · FC 4.99 s
$2 \to 6$: pyAmpliCol only, no provider exists at $2 \to 6$, so no ratio, no gate and no colour · engine 2f3f76d9 (work/v2-mf-8g-3), native loop-native-mf-2to6-2f3f76d9-lto (2bae17f9…), an ancestor of the engine above, not re-run on it · shared host, indicative · internal checks only				
$2 \to 6$	$gg \to gggggg$ 8 legs · QCD · 47 348 families · $N_f = 5$, no top **LC** ε^{-2} ≤ 1.0e−11 · E ≤ 1.4e−3 (2 pts above 10^{-5}) · binary64 = DD to 7.4e−5 · rescue 11/16 · NGluon DD to 6.3e−13 · **FC** ε^{-2} ≤ 8.4e−11 · E ≤ 5.1e−3 (2 pts above 10^{-5}) · Catani ε^{-1} ≤ 7.1e−5 · binary64 = DD to 1.2e−4 · rescue 11/16 Ward FC not run, Ward LC not asserted	**55.3 min** FC 74.0 GiB peak · plan 486 MB · store 25.8 min artifact 0.53 GB · load 4.21 s (1.06 GiB) **LC** 46.1 min · 63.9 GiB · artifact 0.364 GB	**201 s** 1 block of 16 $B = 16$ no provider: no gate inputs 55 % peak 40.1 GiB	**98.0 s** 1 block of 128 $B = 128$ numerator 54 % peak 27.2 GiB
	$u\bar{u} \to gggggg$ 8 legs · QCD · 47 348 families · $N_f = 5$, no top **LC** ε^{-2} ≤ 4.6e−12 · E ≤ 6.2e−7 · binary64 = DD to 8.4e−11 · rescue 1/16 · Ward ≤ 6.5e−12 · **FC** ε^{-2} ≤ 1.1e−11 · E ≤ 5.9e−6 · Catani ε^{-1} ≤ 3.3e−8 · binary64 = DD to 1.7e−10 · rescue 2/16 · Ward ≤ 2.1e−11	**73.6 min** FC 92.9 GiB peak · plan 814 MB · store 449 s artifact 0.143 GB · load 3.85 s (1.66 GiB) **LC** 20.5 min · 47.0 GiB · artifact 0.047 GB	**62.1 s** 1 block of 128 $B = 128$ no provider: no gate numerator 77 % peak 31.7 GiB	**10.9 s** 1 block of 128 $B = 128$ numerator 60 % peak 12.6 GiB
	$u\bar{u} \to u\bar{u}gggg$ 8 legs · QCD · 40 454 families · $N_f = 5$, no top **LC** ε^{-2} ≤ 6.7e−14 · E ≤ 7.5e−7 · binary64 = DD to 1.3e−11 · rescue 2/16 · Ward ≤ 4.3e−12 · **FC** ε^{-2} ≤ 2.0e−13 · E ≤ 6.5e−6 · Catani ε^{-1} ≤ 6.6e−11 · binary64 = DD to 1.3e−11 · rescue 4/16 · Ward ≤ 8.5e−12 (DD 1.1e−28)	**29.8 min** FC 42.5 GiB peak · plan 315 MB · store 46.4 s artifact 0.038 GB · load 1.51 s (0.68 GiB) **LC** 506 s · 18.2 GiB · artifact 0.009 GB	**19.2 s** 1 block of 128 $B = 128$ no provider: no gate numerator 75 % peak 36.0 GiB	**2.67 s** 1 block of 128 $B = 128$ numerator 55 % peak 7.38 GiB

Table 2: **Loop-induced processes** ($|A_{\text{loop}}|^2$; the equivalent of Table 1). How to read it: after Table 3.

	Process · validation	Generation	$\lvert A_{\text{loop}}\rvert^2$ · per point, batched, backend auto serves
pyAmpliCol: engine e852f448, native loop-native-v2-slice-e852f448 (c3fda0c4…) on itphlies (packages landed: lit-amp, lit-r2, lit-ew) · default backend TIRust · providers: lit-ref's production timing records on itphlies · 2026-11-08 · nothing estimated			
	legs · corrections · PLAN §4.1 ids; largest deviation per reference (lit-val)	generate + filter + plan export + probes (share), peak memory; provider generation	median per point at $B = 128$; provider times (lit-ref); gate (lit-perf) or rule; backend (request) · the other backend
$2 \to 1 \cdot 1 \to 2$	$gg \to h$ 3 legs · QCD · LI-A1 validation pending (lit-val)	**6.06 s** incl. probes 3.10 s (51 %) 0.56 GiB peak · plan 0.0100 MB	**0.203 ms** ML · OL · Rc: no provider timing record of the row in li_campaign.json $B = 128$ no gate (2→1) **NinjaRS per amplitude** (auto) · TIRust 0.192 ms (128/128)
	$h \to gg$ 3 legs · QCD · LI-A2 validation pending (lit-val)	**6.03 s** incl. probes 3.08 s (51 %) 0.56 GiB peak · plan 0.0100 MB	**0.216 ms** OL unavailable: decay: the designated OpenLoops ls libraries are 2 -> n scattering libraries · ML · Rc not timed: no production timing yet $B = 128$ no gate (1→2) **NinjaRS per amplitude** (auto) · TIRust 0.205 ms (128/128)
	$h \to aa$ 3 legs · QED · LI-A3, LI-E1 validation pending (lit-val)	**146 s** incl. probes 73.7 s (51 %) 3.23 GiB peak · plan 0.0270 MB	**0.639 ms** OL unavailable: decay: the designated OpenLoops ls libraries are 2 -> n scattering libraries · ML · Rc not timed: no production timing yet $B = 128$ no gate (1→2) **NinjaRS per amplitude** (auto) · TIRust 0.620 ms (128/128)
	$h \to za$ 3 legs · QED · LI-A4, LI-E1 validation pending (lit-val)	**254 s** incl. probes 179 s (71 %) 4.49 GiB peak · plan 0.0350 MB	**0.786 ms** OL unavailable: decay: the designated OpenLoops ls libraries are 2 -> n scattering libraries · ML · Rc not timed: no production timing yet $B = 128$ no gate (1→2) **NinjaRS per amplitude** (auto) · TIRust 0.684 ms (128/128)
	$aa \to h$ 3 legs · QED · LI-E2 validation pending (lit-val)	**146 s** incl. probes 73.6 s (50 %) 3.25 GiB peak · plan 0.0270 MB	**0.614 ms** OL unavailable: no designated OpenLoops loop-squared (ls) library serves this process (PLAN §4.1, Q34 = (b)) · ML · Rc not timed: no production timing yet $B = 128$ no gate (2→1) **NinjaRS per amplitude** (auto) · TIRust 0.598 ms (128/128)
$2 \to 2$	$gg \to aa$ 4 legs · QCD · LI-A5 validation pending (lit-val)	**11.0 s** incl. probes 6.94 s (63 %) 0.67 GiB peak · plan 0.0210 MB	**3.42 ms** ML · OL · Rc: no provider timing record of the row in li_campaign.json $B = 128$ no gate (2→2) **NinjaRS per amplitude** (auto) · TIRust 2.17 ms (128/128)
	$gg \to zz$ 4 legs · QCD · LI-21 validation pending (lit-val)	**13.6 s** incl. probes 9.06 s (67 %) 0.70 GiB peak · plan 0.0290 MB	**8.61 ms** ML · OL · Rc: no provider timing record of the row in li_campaign.json $B = 128$ no gate (2→2) **NinjaRS per amplitude** (auto) · TIRust 5.43 ms (128/128)
	$gg \to w^+w^-$ 4 legs · QCD · LI-22 validation pending (lit-val)	**15.4 s** incl. probes 10.4 s (68 %) 0.71 GiB peak · plan 0.0420 MB	**9.55 ms** ML · OL · Rc: no provider timing record of the row in li_campaign.json $B = 128$ no gate (2→2) **NinjaRS per amplitude** (auto) · TIRust 4.53 ms (128/128)
	$gg \to za$ 4 legs · QCD · LI-23 validation pending (lit-val)	**13.4 s** incl. probes 8.54 s (64 %) 0.69 GiB peak · plan 0.0360 MB	**4.88 ms** ML · OL · Rc: no provider timing record of the row in li_campaign.json $B = 128$ no gate (2→2) **NinjaRS per amplitude** (auto) · TIRust 2.79 ms (128/128)
	$gg \to hh$ 4 legs · QCD · LI-24, LI-A7 validation pending (lit-val)	**8.11 s** ×95 incl. probes 4.92 s (61 %) 0.60 GiB peak · plan 0.0100 MB Rc 85.1 ms	**0.558 ms** ×0.81 **ML 0.692 ms** · Rc 1.36 ms · OL 1.56 ms $B = 128$ no gate (2→2) **NinjaRS per amplitude** (auto) · TIRust 0.404 ms (128/128)
	$gg \to hz$ 4 legs · QCD · LI-25 validation pending (lit-val)	**11.6 s** incl. probes 6.12 s (53 %) 0.68 GiB peak · plan 0.0280 MB	**1.28 ms** ML · OL · Rc: no provider timing record of the row in li_campaign.json $B = 128$ no gate (2→2) **NinjaRS per amplitude** (auto) · TIRust 0.956 ms (128/128)
	$gg \to hg$ 4 legs · QCD · LI-26, LI-A7 validation pending (lit-val)	**7.75 s** incl. probes 4.45 s (57 %) 0.60 GiB peak · plan 0.0110 MB	**1.45 ms** ML · OL · Rc: no provider timing record of the row in li_campaign.json $B = 128$ no gate (2→2) **NinjaRS per amplitude** (auto) · TIRust 0.908 ms (128/128)
	$u\bar{u} \to hg$ 4 legs · QCD · LI-26, LI-A7 validation pending (lit-val)	**6.86 s** incl. probes 3.51 s (51 %) 0.57 GiB peak · plan 0.0100 MB	**0.264 ms** ML · OL · Rc: no provider timing record of the row in li_campaign.json $B = 128$ no gate (2→2) **NinjaRS per amplitude** (auto) · TIRust 0.248 ms (128/128)
	$ug \to hu$ 4 legs · QCD · LI-26 validation pending (lit-val)	**7.00 s** incl. probes 3.52 s (50 %) 0.57 GiB peak · plan 0.0100 MB	**0.258 ms** ML · OL · Rc: no provider timing record of the row in li_campaign.json $B = 128$ no gate (2→2) **NinjaRS per amplitude** (auto) · TIRust 0.249 ms (128/128)
	$gg \to zg$ 4 legs · QCD · LI-27 validation pending (lit-val)	**16.4 s** incl. probes 11.6 s (71 %) 0.70 GiB peak · plan 0.0290 MB	**13.2 ms** ML · OL · Rc: no provider timing record of the row in li_campaign.json $B = 128$ no gate (2→2) **NinjaRS per amplitude** (auto) · TIRust 5.90 ms (128/128)
	$gg \to ag$ 4 legs · QCD · LI-27 validation pending (lit-val)	**15.0 s** incl. probes 10.4 s (69 %) 0.69 GiB peak · plan 0.0230 MB	**6.88 ms** ML · OL · Rc: no provider timing record of the row in li_campaign.json $B = 128$ no gate (2→2) **NinjaRS per amplitude** (auto) · TIRust 3.43 ms (128/128)

	Process · validation	Generation	$\lvert A_{\text{loop}}\rvert^2$ · per point, batched, backend auto serves
	pyAmpliCol: engine e852f448, native loop-native-v2-slice-e852f448 (c3fda0c4…) on itphlies (packages landed: lit-amp, lit-r2, lit-ew) · default backend TIRust · providers: lit-ref's production timing records on itphlies · 2026-11-08 · nothing estimated		
	legs · corrections · PLAN §4.1 ids; largest deviation per reference (lit-val)	generate + filter + plan export + probes (share), peak memory; provider generation	median per point at $B = 128$; provider times (lit-ref); gate (lit-perf) or rule; backend (request) · the other backend
1 → 3	$e^+e^- \to ha$ 4 legs · QED · LI-A4, LI-E3 validation pending (lit-val)	**248 s** incl. probes 127 s (51 %) 5.65 GiB peak · plan 0.0670 MB	**1.14 ms** OL unavailable: no designated OpenLoops loop-squared (ls) library serves this process (PLAN §4.1, Q34 = (b)) · ML · Rc not timed: no production timing yet $B = 128$ no gate (2→2) **NinjaRS per amplitude** (auto) · TIRust 1.22 ms (128/128)
	$aa \to aa$ 4 legs · QED · LI-A6, LI-E2 validation pending (lit-val)	**227 s** incl. probes 115 s (51 %) 6.11 GiB peak · plan 0.0390 MB	**7.50 ms** OL unavailable: no designated OpenLoops loop-squared (ls) library serves this process (PLAN §4.1, Q34 = (b)) · ML · Rc not timed: no production timing yet $B = 128$ no gate (2→2) **NinjaRS per amplitude** (auto) · TIRust 4.05 ms (128/128)
	$aa \to zz$ 4 legs · QED · LI-E2 validation pending (lit-val)	**432 s** incl. probes 305 s (71 %) 6.83 GiB peak · plan 0.0920 MB	**18.7 ms** OL unavailable: no designated OpenLoops loop-squared (ls) library serves this process (PLAN §4.1, Q34 = (b)) · ML · Rc not timed: no production timing yet $B = 128$ no gate (2→2) **NinjaRS per amplitude** (auto) · TIRust 10.2 ms (128/128)
	$aa \to hh$ 4 legs · QED · LI-E2 validation pending (lit-val)	**186 s** incl. probes 97.4 s (52 %) 3.66 GiB peak · plan 0.0340 MB	**1.72 ms** OL unavailable: no designated OpenLoops loop-squared (ls) library serves this process (PLAN §4.1, Q34 = (b)) · ML · Rc not timed: no production timing yet $B = 128$ no gate (2→2) **NinjaRS per amplitude** (auto) · TIRust 1.33 ms (128/128)
	$e^+e^- \to hh$ 4 legs · QED · LI-E3 validation pending (lit-val)	**239 s** incl. probes 123 s (52 %) 5.63 GiB peak · plan 0.0390 MB	**0.988 ms** OL unavailable: no designated OpenLoops loop-squared (ls) library serves this process (PLAN §4.1, Q34 = (b)) · ML · Rc not timed: no production timing yet $B = 128$ no gate (2→2) **NinjaRS per amplitude** (auto) · TIRust 1.03 ms (128/128)
	$h \to ggg$ 4 legs · QCD · LI-37 validation pending (lit-val)	**7.90 s** incl. probes 4.59 s (58 %) 0.60 GiB peak · plan 0.0110 MB	**1.58 ms** OL unavailable: decay: the designated OpenLoops ls libraries are 2 -> n scattering libraries · ML · Rc not timed: no production timing yet $B = 128$ no gate (1→3) **NinjaRS per amplitude** (auto) · TIRust 1.03 ms (128/128)
	$z \to ggg$ 4 legs · QCD · LI-37 validation pending (lit-val)	**15.5 s** incl. probes 10.7 s (69 %) 0.70 GiB peak · plan 0.0290 MB	**9.99 ms** OL unavailable: decay: the designated OpenLoops ls libraries are 2 -> n scattering libraries · ML · Rc not timed: no production timing yet $B = 128$ no gate (1→3) **NinjaRS per amplitude** (auto) · TIRust: 42/128 failed closed, ruling (d)
2 → 3	$e^+e^- \to ggg$ 5 legs · QED · LI-E4 validation pending (lit-val)	**118 s** ×739 incl. probes 57.9 s (49 %) 3.65 GiB peak · plan 0.0380 MB Rc 160 ms	**8.07 ms** ×1.8 **Rc 4.44 ms** · ML 15.5 ms · OL unavailable: no designated OpenLoops loop-squared (ls) library serves this process (PLAN §4.1, Q34 = (b)) $B = 128$ no verdict yet **NinjaRS per amplitude** (auto) · TIRust: failed closed on every point (128/128), ruling (d)
	$gg \to hhh$ 5 legs · QCD · LI-31 validation pending (lit-val)	**10.6 s** ×137 incl. probes 7.21 s (68 %) 0.68 GiB peak · plan 0.0120 MB Rc 77.1 ms	**3.82 ms** ×0.68 **Rc 5.64 ms** · ML 9.03 ms · OL 22.9 ms $B = 128$ <5 ms: not gated **NinjaRS per amplitude** (auto) · TIRust: 42/128 failed closed, ruling (d)
	$gg \to hgg$ 5 legs · QCD · LI-32, LI-A7 validation pending (lit-val)	**29.4 s** ×140 incl. probes 25.2 s (86 %) 0.77 GiB peak · plan 0.0190 MB Rc 210 ms	**43.2 ms** ×1.7 **Rc 25.5 ms** · OL 25.7 ms · ML 86.2 ms $B = 128$ no verdict yet **NinjaRS per amplitude** (auto) · TIRust: failed closed on every point (128/128), ruling (d)
	$gg \to zzg$ 5 legs · QCD · LI-33 validation pending (lit-val)	**195 s** ×126 incl. probes 189 s (97 %) 1.17 GiB peak · plan 0.0380 MB Rc 1.55 s	**367 ms** ×2.1 **Rc 178 ms** · OL 183 ms · ML 721 ms $B = 128$ no verdict yet **NinjaRS per amplitude** (auto) · TIRust: failed closed on every point (128/128), ruling (d)
	$gg \to aag$ 5 legs · QCD · LI-33 validation pending (lit-val)	**118 s** ×429 incl. probes 113 s (96 %) 0.99 GiB peak · plan 0.0300 MB Rc 274 ms	**157 ms** ×3.3 **Rc 47.7 ms** · OL 52.5 ms · ML 200 ms $B = 128$ no verdict yet **NinjaRS per amplitude** (auto) · TIRust: failed closed on every point (128/128), ruling (d)
	$gg \to w^+w^-g$ 5 legs · QCD · LI-33 validation pending (lit-val)	**187 s** ×136 incl. probes 180 s (97 %) 1.24 GiB peak · plan 0.0580 MB Rc 1.37 s	**372 ms** ×5.4 **Rc 68.8 ms** · OL 84.4 ms · ML 347 ms $B = 128$ no verdict yet **NinjaRS per amplitude** (auto) · TIRust: failed closed on every point (128/128), ruling (d)
	$gg \to zzz$ 5 legs · QCD · LI-34 validation pending (lit-val)	**147 s** ×170 incl. probes 131 s (89 %) 1.06 GiB peak · plan 0.0330 MB Rc 862 ms	**340 ms** ×2.5 **Rc 134 ms** · OL 151 ms · ML 560 ms $B = 128$ no verdict yet **NinjaRS per amplitude** (auto) · TIRust: 85/128 failed closed, ruling (d)
	$gg \to hzg$ 5 legs · QCD · LI-35 validation pending (lit-val)	**34.0 s** ×172 incl. probes 28.6 s (84 %) 0.82 GiB peak · plan 0.0330 MB Rc 198 ms	**47.6 ms** ×2.5 **Rc 19.1 ms** · OL 32.3 ms · ML 73.9 ms $B = 128$ no verdict yet **NinjaRS per amplitude** (auto) · TIRust: 42/128 failed closed, ruling (d)

	Process · validation	Generation	$\lvert A_{\text{loop}}\rvert^2$ · per point, batched, backend auto serves
	pyAmpliCol: engine e852f448, native loop-native-v2-slice-e852f448 (c3fda0c4…) on itphlies (packages landed: lit-amp, lit-r2, lit-ew) · default backend TIRust · providers: lit-ref's production timing records on itphlies · 2026-11-08 · nothing estimated		
	legs · corrections · PLAN §4.1 ids; largest deviation per reference (lit-val)	generate + filter + plan export + probes (share), peak memory; provider generation	median per point at $B = 128$; provider times (lit-ref); gate (lit-perf) or rule; backend (request) · the other backend
$2 \to 4$	$gg \to agg$ 5 legs · QCD · LI-35 validation pending (lit-val)	**199 s** ×272 incl. probes 193 s (97 %) 1.34 GiB peak · plan 0.0460 MB Rc 733 ms	**291 ms** ×3.8 **Rc 75.8 ms** · OL 78.8 ms · ML 250 ms $B = 128$ no verdict yet **NinjaRS per amplitude** (auto) · TIRust: failed closed on every point (128/128), ruling (d)
$2 \to 4$	$gg \to zgg$ 5 legs · QCD · LI-35 validation pending (lit-val)	**199 s** ×137 incl. probes 192 s (97 %) 1.47 GiB peak · plan 0.0520 MB Rc 1.45 s	**453 ms** ×3.9 **Rc 116 ms** · ML 476 ms · OL unavailable: no OpenLoops ls library serves g g > z g g: the public channel database of OpenLoops 2.1.5 (proclib/channels_public.rinfo, 2026-11-06) has zgg/zggg (ppvj2) but no zgggg channel; row ML/RC only (PLAN §3.2) $B = 128$ no verdict yet **NinjaRS per amplitude** (auto) · TIRust: failed closed on every point (128/128), ruling (d)
$2 \to 4$	$u\bar{u} \to hgg$ 5 legs · QCD · LI-36 validation pending (lit-val)	**10.4 s** ×14 incl. probes 5.88 s (57 %) 0.61 GiB peak · plan 0.0110 MB Rc 727 ms	**2.35 ms** ×1.5 **Rc 1.53 ms** · OL 2.61 ms · ML 2.63 ms $B = 128$ <5 ms: not gated **NinjaRS per amplitude** (auto) · TIRust: 43/128 failed closed, ruling (d)
$2 \to 4$	$u\bar{u} \to d\bar{d}h$ 5 legs · QCD · LI-36 validation pending (lit-val)	**9.11 s** ×37 incl. probes 4.39 s (48 %) 0.58 GiB peak · plan 0.0100 MB Rc 244 ms	**0.511 ms** ×3.8 **ML 0.134 ms** · OL 0.271 ms · Rc 0.536 ms $B = 128$ <5 ms: not gated **NinjaRS per amplitude** (auto) · TIRust 0.604 ms (128/128)
$2 \to 4$	$gg \to hhgg$ 6 legs · QCD · LI-41 validation pending (lit-val)	**397 s** ×1044 incl. probes 388 s (98 %) 1.77 GiB peak · plan 0.0470 MB Rc 380 ms	**981 ms** ×5.2 **Rc 189 ms** · OL 572 ms · ML 671 ms $B = 128$ no verdict yet **NinjaRS per amplitude** (auto) · TIRust: failed closed on every point (128/128), ruling (d)
$2 \to 4$	$gg \to hggg$ 6 legs · QCD · LI-42 validation pending (lit-val)	**14.4 min** ×269 incl. probes 14.2 min (99 %) 5.11 GiB peak · plan 0.100 MB Rc 3.20 s	**3.37 s** ×5.8 **Rc 577 ms** · OL 853 ms · ML 1.98 s $B = 128$ no verdict yet **NinjaRS per amplitude** (auto) · TIRust: failed closed on every point (128/128), ruling (d)
$2 \to 4$	$gg \to e^+e^-\mu^+\mu^-$ 6 legs · QCD · LI-43 validation pending (lit-val)	**14.3 s** ×68 incl. probes 8.30 s (58 %) 0.70 GiB peak · plan 0.0380 MB Rc 209 ms	**4.72 ms** ×1.3 **Rc 3.70 ms** · OL 6.64 ms · ML 33.8 ms $B = 128$ Rc match open (lit-perf): ×0.71 to OL without it <5 ms: not gated **NinjaRS per amplitude** (auto) · TIRust 3.21 ms (128/128)
$2 \to 4$	$gg \to aagg$ 6 legs · QCD · LI-44 validation pending (lit-val)	***not run*** generation beyond the 45-min rule (R9): generate + plan export + exact helicity-zero filter done in 14.0 s (peak 0.63 GiB); the mandatory double-double generation-time probes still running when stopped at 45 min 44 s (2.9 GB RSS; 23:03 UTC); no time **Rc 1.67 s** · OL 1.96 s · ML 5.89 s	
$2 \to 4$	$gg \to zzzz$ 6 legs · QCD · LI-44 validation pending (lit-val)	***not run*** generation beyond the 45-min rule (R9): generate + plan export + exact helicity-zero filter done in 9.8 s (peak 0.60 GiB); the mandatory double-double generation-time probes still running when stopped at 45 min 45 s (3.7 GB RSS; 23:03 UTC); no time **Rc 5.07 s** · ML 19.9 s · OL unavailable: no OpenLoops ls library serves g g > z z z z (public channel database of OpenLoops 2.1.5, 2026-11-06); row ML/RC only (PLAN §3.2)	
$2 \to 5$	$gg \to e^+e^-\mu^+\mu^-g$ or $gg \to hgggg$ 7 legs · QCD · LI-51 validation pending (lit-val)	**94.8 s** ×81 incl. probes 87.1 s (92 %) 1.14 GiB peak · plan 0.0530 MB Rc 1.17 s	**195 ms** ×3.1 **Rc 63.4 ms** · OL 252 ms · ML 1.20 s $B = 128$ Rc match open (lit-perf): ×0.77 to OL without it no verdict yet **NinjaRS per amplitude** (auto) · TIRust 81.0 ms (128/128)
EFT	rank > N+2 EFT row (HC_NLO_X0 or SMEFT) EFT · LI-X1 · TIRust only validation pending (lit-val)	***pending*** served by lit-eft (not landed at this head) ML: no provider timing record of the row in li_campaign.json	



Not drawn: 25 of the 42 rows of Table 2 (3 no pyAmpliCol time at B = 128, 22 no matched provider time). Drawn with a provider whose match is open (lit-perf; its values no reference): $gg \to e^+e^-\mu^+\mu^-$ (Recola), $gg \to e^+e^-\mu^+\mu^-g$ (Recola).

Figure 1: **Batched full colour against the gate** (finite part, $B = 128$; the rows of Table 1 with a matched provider and, as diamonds, the rows of Table 2 with a pyAmpliCol time and a matched provider time).

Table 3: **Gate and stage split at $B = 128$**, ms per point: the numbers behind the gate chips and stage bars of Table 1 and behind Figure 1. pyAmpliCol's median (full colour), the binding limit and its provider, the uncertainty band of time ÷ limit, the gate chip of Table 1, and pyAmpliCol's time per stage.

Process	ms/pt	Limit	Band	Gate	inputs	numerator	scatter	NinjaRS	masters	glue
$gg \to gg$	**1.29**	—	—	no gate (2→2)	0.191	0.665	0.0663	0.113	0.0372	0.219
$u\bar{u} \to d\bar{d}g$	**1.72**	1.97 OL	0.862–0.884	<5 ms: not gated ×0.87	0.0768	0.647	0.397	0.142	0.109	0.342
$u\bar{u} \to u\bar{u}g$	**2.64**	3.80 OL	0.678–0.711	<5 ms: not gated ×0.69	0.126	1.10	0.424	0.254	0.190	0.525
$u\bar{u} \to e^-e^+g$	**0.788**	0.948 ML	0.830–0.832	<5 ms: not gated ×0.83	0.0516	0.285	0.129	0.0355	0.0262	0.263
$u\bar{u} \to ggg$	**4.34**	16.1 OL	0.259–0.280	<5 ms: not gated ×0.27	0.389	2.32	0.411	0.456	0.211	0.576
$gg \to ggg$	**20.1**	126 OL	0.155–0.164	pass ×0.16	5.57	9.88	0.518	2.27	0.524	1.14
$u\bar{u} \to e^-e^+gg$	**4.21**	13.3 Rc	0.314–0.319	<5 ms: not gated ×0.32	0.161	2.26	0.589	0.435	0.237	0.524
$gg \to t\bar{t}gg$	**79.8**	613 OL	0.128–0.133	pass ×0.13	11.0	34.9	0.888	18.9	6.63	7.49
$u\bar{u} \to gggg$	**65.3**	486 OL	0.130–0.139	pass ×0.13	9.37	37.2	1.74	9.04	2.98	4.78
$gg \to gggg$	**307**	3 690 OL	0.0824–0.0838	pass ×0.083	93.7	158	2.33	38.4	6.05	8.29
$u\bar{u} \to e^-e^+d\bar{d}g$	**13.9**	45.9 Rc	0.278–0.329	pass ×0.30	0.570	6.62	1.79	1.90	1.27	1.58
$u\bar{u} \to e^-e^+ggg$	**47.9**	326 Rc	0.135–0.159	pass ×0.15	1.56	30.1	2.05	7.93	2.75	3.53
$gg \to t\bar{t}e^-e^+g$	**51.1**	525 OL	0.0897–0.105	pass ×0.097	2.88	29.0	0.734	10.2	4.26	4.07
$gg \to t\bar{t}e^-e^+g$ (EW)	**131**	1 280 Rc	0.0935–0.111	pass ×0.10	6.29	60.7	1.75	29.7	16.5	16.2
$u\bar{u} \to ggggg$	**1 310**	569 000 ML	0.00210–0.00248	pass ×0.0023	215	804	9.27	186	46.0	46.8
$gg \to ggggg$	**4 990**	—	—	no gate (no matched provider)	1 970	2 580	11.0	344	42.0	28.4

Reading the tables

Table 1 (Born-interfered one-loop processes) has one row per test process, by multiplicity. A cell gives pyAmpliCol's value in bold and, at its right, the multiplier *pyAmpliCol ÷ the fastest provider* on this host (in bold among the provider values). A provider time marked ≤ ($u\bar{u} \to ggggg$) comes from a run outside the protocol that does the production work and more (Recola's reference run evaluates the IR poles too), so it bounds that provider's production time from above, and the multiplier against it is a lower bound (≥); it is never a gate limit. Only run times are coloured: green faster, white at parity, red slower ($\log_2$ scale, clamped at ×1/16 and

×16). Generation multipliers are uncoloured (generation time is not the main metric) and leading colour (LC) has no provider ratio. Grey cells say why a value is missing. Run times are medians per phase-space point at $B = 128$, finite part, binary64, one pinned thread, stability checks off; generation is generate + plan export (model loading excluded) with its peak memory and plan size. The providers run in production mode: MadLoop 3.7.0 (ML), Recola 2.2.5 with COLLIER (Rc), OpenLoops 2.1.5 from prebuilt libraries (OL: no generation time). Each cell shows the most recent measured configuration (`scripts/build_summary.py`, by git ancestry of the revisions), named once in the table's first row (engine, native build and digest, date, host); a cell from another configuration names its origin.

Gate chips (Table 1, Figure 1, Table 3): the time over the binding limit, the smallest of MadLoop's time and 1.30 × each of OpenLoops' and Recola's; **pass** or **fail** when the whole uncertainty band (Table 3) lies below or above 1, else **open**. A process below 5 ms per point has no blocking gate ("<5 ms: not gated", user rule of 2026-10-24) and its chip keeps the measured ratio; $2 \to 2$ has no gate. The gate is taken at $B = 128$ only.

Stage bars (the full-colour cell names the largest share; Table 3 gives every stage in ms): inputs = trees, Born and counterterm inputs; numerator = seeds, fused factors and the dressing of the open-loop objects; scatter = closures into NinjaRS lanes; NinjaRS = the reduction without the masters; masters = the master integrals (canonical arguments, cache, oneloopmaster); glue = point parameters, input conversion, assembly and the rest of the call.

Validation (process cell): the largest deviation $|a - b| \,/\, \max(|a|, |b|, |B|\ \alpha_s/2\pi)$ of the binary64 finite part per reference at the timed points (NJet DD: NJet 3.1.2 in double-double), and the median over the points and references; † the largest lies within pyAmpliCol's binary64 stability estimate at that point; ¶ it is the reference's own: pyAmpliCol's double-double value lies as far from it, and its binary64 value lies within its estimate of the double-double one. ‡ $gg \to ggggg$: the reference check's points 4 and 8 (1.5×10^{-7} and 2.6×10^{-8} from NJet DD) exceed its tolerance 2×10^{-8} in binary64: **accepted (user 2026-10-25): binary64 within the estimate, DD = NJet** (estimates E 1.5×10^{-6} and 3.4×10^{-7}; pyAmpliCol's double-double values equal the reference's); the check keeps its tolerance and, asserting double-double at these points, passes.

Notes under a value qualify it and leave it as measured, in amber a known limitation, in grey its run conditions: $gg \to t\bar{t}e^-e^+g$, full colour ("model: the workspace's py3 G_mu copy"): timed on reference/models/loop_qcd_qed_sm_Gmu_py3 (the harness's processes.json default, as at M-SM), not on the shipped patched UFO (UFOmodeldb 6ca4b120 + the user's issue-1 patch) that the M-SM campaign validated; the speed gate is taken on this timing.

The $2 \to 6$ rows (`summary.json` `preliminary`, from `data/2to6.json`) have no provider, so no multiplier, gate or colour; their engine is stated once in the table. Generation = generate + plan export, with the store, artifact and load beside it ("FC" marks the full-colour generation, LC below it); the timed evaluation is binary64, finite part, checks off, one pinned thread, at the largest batch measured. The process cell lists the internal checks (ε^{-2}: double pole against the universal value; E: the stability estimate, with the points above 10^{-5}; rescue: the stream points the default rescue policy's binary64 level sends up; Ward on the external gluons; binary64 against double-double; Catani at full colour; NGluon at leading colour on the all-gluon row) and, in grey, those not run or not asserted.

Table 2 follows Table 1′s rules for loop-induced processes, $|A_{\text{loop}}|^2$ without a Born: one row per process of the M-LIT validation matrix (`m-lit/PLAN.md` §4.1; its ids in the process cell), grouped by multiplicity; the backend-unit, zero, identity, observable and complex-mass-scheme rows (LI-B1, LI-B2, LI-Z1, LI-Z1b, LI-Z2, LI-W1, LI-L1, LI-P1, LI-C1, LI-S1, LI-O1) are validation-only and not listed. The cell is the batched $B = 128$ time per point (median over its timing blocks after an untimed warm-up, finite part, one pinned thread, checks off) of the record the backend `auto` serves at the row's revision: its effective backend in bold, the request in brackets. At this head `auto` takes NinjaRS per amplitude on every timed row, with the runtime's reason (verbatim): "auto: NinjaRS; TIRust, the loop-induced default, serves the summed virtual on explicit request until its agreement with NinjaRS on loop-induced processes is established". The default backend of loop-induced processes, TIRust, serves them on explicit request until then, so the plan's "on TIRust where `lit-tir` serves it" is overtaken (coordinator ruling, slice-status DECISIONS 2026-11-07 21:50 R9). Beside it, the other backend requested explicitly in the same process on the same points: its time and served fraction (128/128) where it served all 128 points of every block, else the points it failed closed per block of 128 with the recorded cause ("ruling (d)": binary64 TIRust serves only the rows its conditioning gates pass, its method choice being off at this head, slice-status DECISIONS 2026-11-07 ~17:20 (d), until `lit-tir-mp`; a call that fails points is timed whole, which is no served-point time); a record with a failed point is never the cell, never coloured and never drawn in Figure 1. The multiplier against the fastest matched provider (MadLoop at its loop-induced production setting 7|6|1, OpenLoops `ls`, Recola; production mode) colours the cell with Table 1′s scale. A provider whose timing record excludes its values as a reference (Recola's on-shell widths against the fixed-width scheme of the others) is timed and shown, but whether it is a matched provider of the gate is `lit-perf`'s call (slice-status DECISIONS 2026-11-06 ~19:06): where it is the fastest, the cell says "match open (lit-perf)" and gives the multiplier to the fastest other provider beside it, uncoloured. Generation, uncoloured, is the full production `generate` cost (model loading excluded): generate + the exact helicity-zero filter + plan export + the generation-time double-double finiteness and Ward probes, which run for every loop-induced process (their time and share beneath the value), with the peak memory over all of them; the provenance line names pyAmpliCol's host and the providers'. Rows of four legs or fewer ($2 \to 1$, $1 \to 2$, $2 \to 2$, $1 \to 3$) are reported without a gate; from $2 \to 3$ upward the gate is Table 1′s (faster than MadLoop and at most 1.30× each matched OpenLoops and Recola time, on TIRust), "<5 ms: not gated" below 5 ms per point, no gate without a matched provider; verdict chips come with the gate tool's records, "no verdict yet" before them ("no provider time yet" while no provider of the row is timed). Grey cells are explicit markers: **pending** (the package that serves the row, named in the cell, has not landed at this head, or the input file that fills the cell does not exist yet), **not run** (a landed package without a record of

the row, or a run stopped before its timing, with its measured progress, e.g. a generation beyond the 45-min rule), **failed closed** or **refused** (the record of the cell's request failed points or was refused: no time), and a provider's reason where it has no matched time. Nothing is estimated or extrapolated; each cell shows the newest record of its row. At this head 41 of 42 rows are served; inputs: `data/li-timing.json` present, `tools/loop_reference/li_campaign.json (provider_timing_records)` present, `data/lit.json` not written yet (lit-val); the registry against `lit-ref`'s manifest (`tools/loop_reference/li_campaign.json`): checked (39 rows, 42 ids).

Figure 1 places each row's batched full-colour value of Table 1 (dots) against the providers (diamonds, the darkest the fastest) and the gate limit (red bar, the shaded side fails; grey bar: the limit of a process below 5 ms, not gated; dotted line: 5 ms), for the rows with a matched provider. Beneath them, under a rule, the rows of Table 2 with a batched pyAmpliCol time and a matched provider time, pyAmpliCol as a larger diamond in its colour, on the same axis with their providers and, from $2 \to 3$ upward, the gate limit (MadLoop's time and 1.30× each other matched provider's, the smallest); the rows without both are not drawn, counted in the key by reason.

Table 3 lists, for every row of Table 1 timed at $B = 128$, the numbers its gate chip and stage bar summarise: the binding limit and its provider, the uncertainty band of time ÷ limit (the standard error of pyAmpliCol's median over its timing blocks plus the provider's run-to-run half-range), and the time of each stage. Data: `data/summary.json` and `data/summary-li.json`, written by `scripts/build_summary.py` from the campaign records in `data/` (`README.md`).